# ViewTouch Label Utility

## User Manual

Shelf, price, cigar and receiving labels for ViewTouch POS

Raspberry Pi label station · Zebra label printers

November 2026



# Contents

1. How the program works . . . . . 3

2. Starting and closing . . . . . 4

3. Screen 1 — Start . . . . . 5

4. Screen 2 — Categories (Tree View) . . . . . 7

5. Screen 3 — Items . . . . . 8

6. Screen 4 — Preview & Print . . . . . 10

7. Printing labels . . . . . 10

8. Editing labels (the label designer) . . . . . 14

9. Importing labels . . . . . 17

10. Your label templates . . . . . 18

11. Updating the program . . . . . 20

12. Troubleshooting . . . . . 20

13. Quick reference . . . . . 22

# 1. How the program works

ViewTouch Label Utility prints shelf, price, cigar-stick and receiving labels for the items in your ViewTouch POS. It runs on the Raspberry Pi next to ViewTouch and sends labels to your Zebra label printers.

### What it does, in order

1. **Reads your items from ViewTouch.** Names, prices, UPCs (including alternate and carton UPCs), brands and categories come straight from ViewTouch's database. The program only *reads* it — nothing you do here changes an item in ViewTouch.
2. **Uses a label template.** A template is a saved label design: its size, the printer it goes to, and where each piece (description, price, barcode, date…) sits on the label.
3. **Draws the whole label as one picture** at the printer's own resolution (203 dots per inch on most Zebras). Text and barcodes are drawn by the program, bar by bar, so every bar is an exact number of printer dots wide.
4. **Sends that picture to the printer.** The printer only prints the dots it is given; it never substitutes its own fonts or barcodes. This is the same way the old ILSA system printed, which is why labels look identical every time.

> **Good to know:** What you see on the print-preview screen is drawn from the same layout as the printout. Double-click the preview to see the exact printer dots.

### Where things are kept

| What | Where |
|---|---|
| **The program** | `~/viewtouch-main/modern/tools/vt_label_utility.py` |
| **Your label templates (live copy)** | `~/.local/share/ViewTouch/claude_label_templates.json` |
| **Starting templates shipped with the program** | `~/viewtouch-main/modern/seed/templates/claude_label_templates.json` |
| **Items, prices, UPCs, printers** | ViewTouch's own database (read only) |
| **Error log** | `/tmp/vt_label_utility.log` (cleared when the Pi restarts) |
| **Last label sent to a printer** | `/tmp/last_label.zpl` and `/tmp/VT_Labels.pdf` |

The label utility keeps its own template list, so templates you delete stay deleted.

# 2. Starting and closing

- **From ViewTouch:** on the Manager screen, press the **Label Utility** button. While the label utility is open, ViewTouch will not log you out for being idle.
- **For one item:** if ViewTouch opens the label utility for a particular item (a button set up with the `openLabelUtility` command and a UPC), it goes straight to that item's label preview. Any of the item's codes works — primary, alternate or carton UPC.
- The program opens full-screen. To close it, press **Close** (red, bottom-left).
- The title bar says *ViewTouch Label Utility v1.0*; the start screen shows how many ViewTouch items it loaded.

> **Note:** Item changes made in ViewTouch (new items, price changes) appear here the next time the program opens, or straight away if you press **Reload viewtouch.db**.

## The four screens

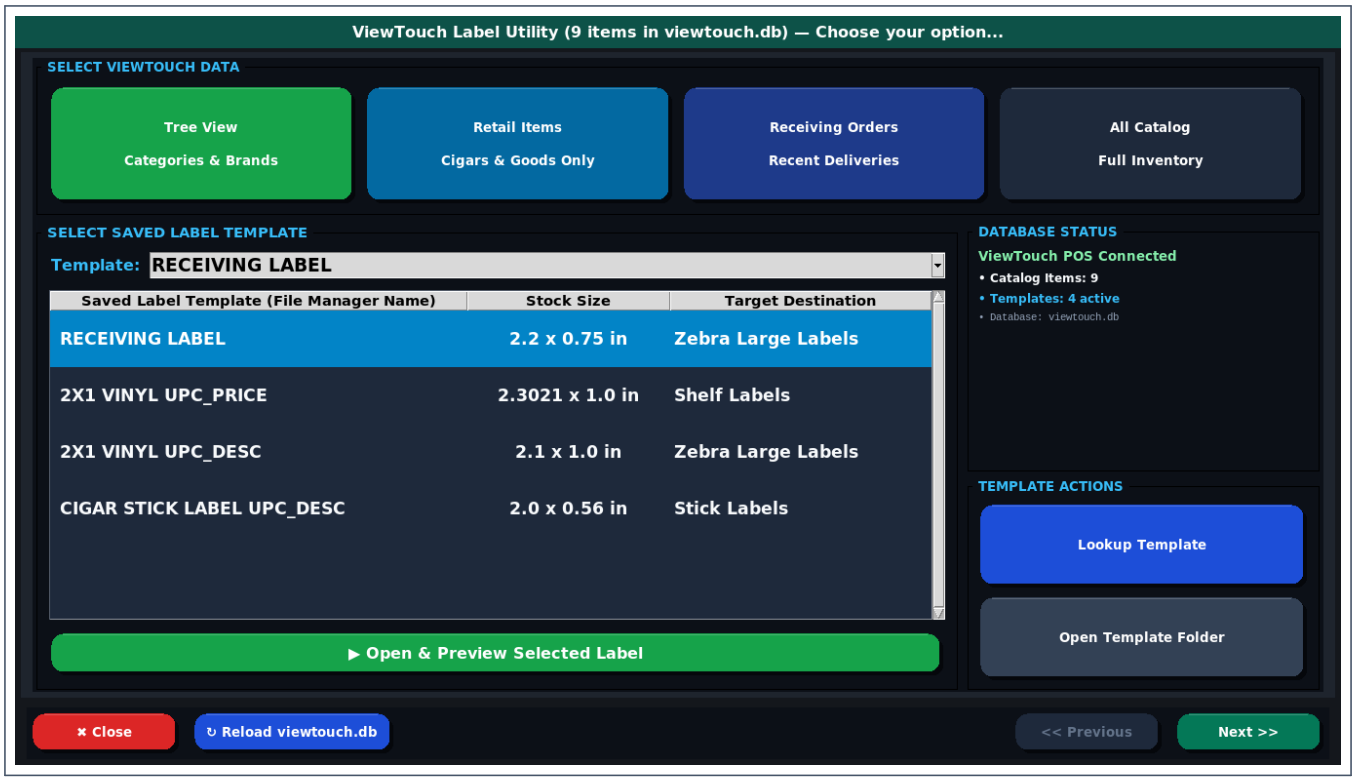
The program works like a wizard. You move forward with **Next >>** and back with **<< Previous** (bottom of the window).

| Screen | What you do there |
|---|---|
| **1. Start** | Choose which items to work with, or jump straight to a saved label template. |
| **2. Categories** | Narrow the items down by group, category or brand (Tree View only). |
| **3. Items** | Tick the items to print, set quantities, and choose which UPC goes on each label. |
| **4. Preview & Print** | Pick the template, check the preview, edit the label design, and print. |

### Bottom bar (always visible)

| Button | What it does |
|---|---|
| **Close** | Closes the program. |
| **Reload viewtouch.db** | Re-reads all items from ViewTouch (use after changing items in ViewTouch). This also resets any UPC choices made on screen 3. |
| **<< Previous** | Goes back one screen. |
| **Next >>** | Goes forward one screen. |

# 3. Screen 1 — Start




Screen 1 — choose items or a saved template

## Select ViewTouch data (top row)

Each big button chooses which items you will work with and moves you on immediately.

| Button | Shows | Goes to |
|---|---|---|
| **Tree View — Categories & Brands** | All items, organised by group, category and brand so you can pick a section. | Screen 2 |
| **Retail Items — Cigars & Goods Only** | Only retail items (items with a real barcode, merchandise, or cigar/tobacco/retail categories). | Screen 3 |
| **Receiving Orders** | Items sold in packs or cartons, plus retail items — handy after a delivery. | Screen 3 |
| **All Catalog — Full Inventory** | Every item. | Screen 3 |

## Select saved label template (left)

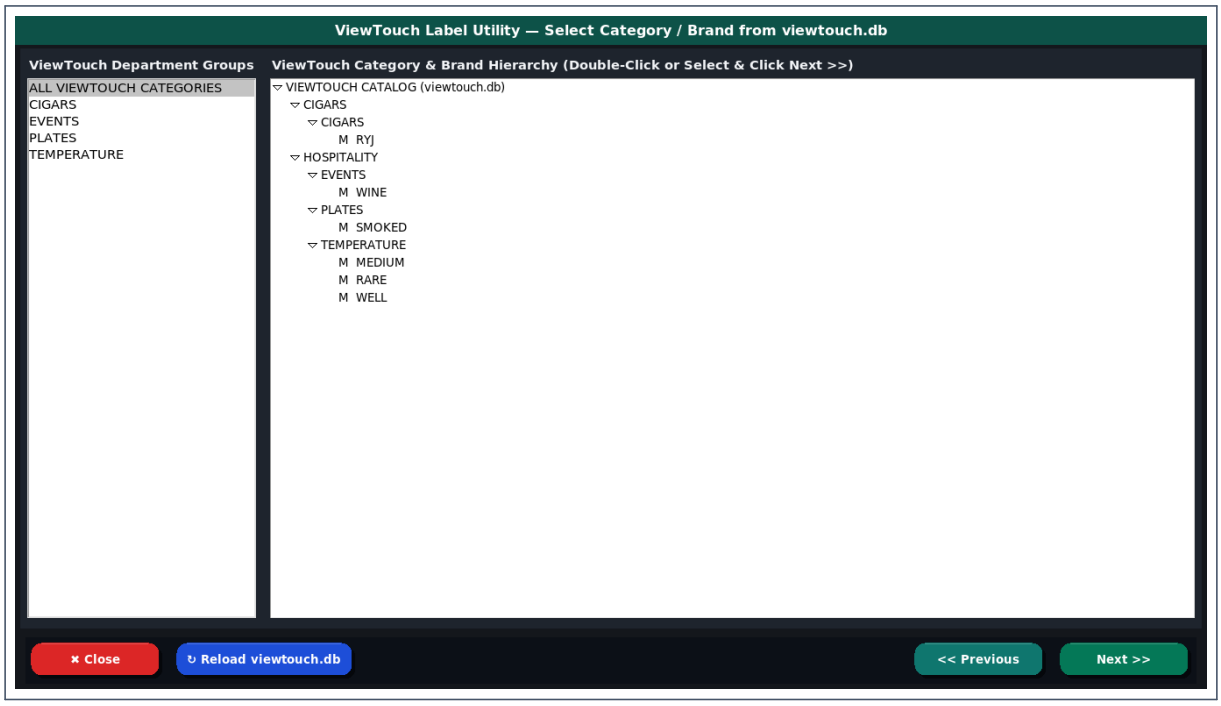
Lists every label template. To go straight to a label's preview:

- Tap a template in the list (or choose it in the **Template** drop-down), then press **Open & Preview Selected Label** — or simply double-click the template.
- This opens screen 4 with that template. If no items have been ticked yet, the preview uses the first item in the list. This is the quickest way to print a **RECEIVING LABEL**, which does not need an item.

## Database status and template actions (right)

| Item | What it means / does |
|---|---|
| **Database status** | Confirms ViewTouch is connected and shows how many items and templates were loaded. |
| **Lookup Template** | Imports a label template from a file — including ILSA **.rtm** labels. See section 9. |
| **Open Template Folder** | Opens the `seed/templates` folder in the file manager — a handy place to keep files you want to import. |

# 4. Screen 2 — Categories (Tree View)

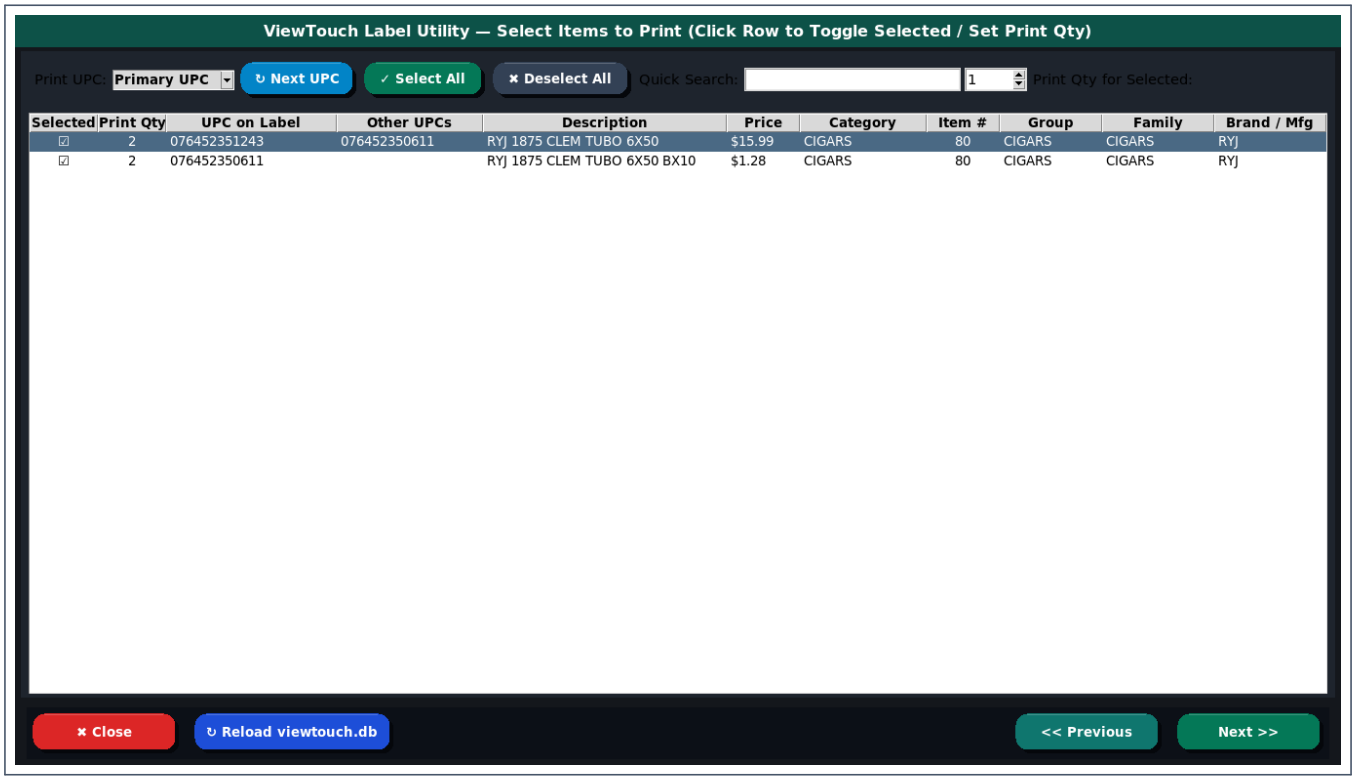



Screen 2 — groups, categories and brands

- **Left list — ViewTouch Department Groups:** tap a category to show only that category in the tree. **ALL VIEWTOUCH CATEGORIES** shows everything.
- **Right — Category & Brand Hierarchy:** groups (Cigars, Retail…) contain categories, which contain brands (shown with an **M** in front).
- Tap a group, category or brand and press **Next >>** — or just double-click it — to list only those items on screen 3. Selecting the top line (or nothing) lists every item.

# 5. Screen 3 — Items

This screen lists the items. Tick the ones you want labels for.




Screen 3 — two items ticked; the highlighted row will print its primary UPC

## Selecting items

- **Tap a row** to tick it (the empty box in Selected gets a tick). Tap again to untick. Ticking a row also sets its **Print Qty** to the number in **Print Qty for Selected**.
- **Select All** / **Deselect All** tick or untick every item in the list.
- **Quick Search** filters the list as you type. It matches description, brand, category and *any* of an item's UPCs — so you can scan or type an alternate UPC to find an item.
- **Print Qty for Selected** (right) sets the quantity on every ticked item.
- Press **Next >>** to go to the preview with the ticked items. If nothing is ticked, the first item is used.

## Columns

| Column | Meaning |
|---|---|
| **Selected / Print Qty** | Whether the item is ticked and its quantity. |
| **UPC on Label** | The code that will be printed. If it is not the primary UPC, it is tagged, e.g. *04252614 [Alt UPC 1]*. |
| **Other UPCs** | The item's other codes you could print instead. |
| **Description, Price, Category, Item #, Group, Family, Brand / Mfg** | Item details from ViewTouch. |

## Choosing which UPC goes on the label

Many items have more than one code in ViewTouch — a primary UPC, alternate UPCs, and for packs the carton UPC. You can print any of them. **This only affects the label; it never changes the item in ViewTouch.**

| Control | What it does |
|---|---|
| **Print UPC (drop-down)** | Switches the ticked items (or every listed item if none are ticked) to **Primary UPC**, **Alt UPC 1–3** or **Carton UPC**. Items that don't have that code keep their primary UPC. |
| **Next UPC** | Steps the highlighted row (or the ticked rows) to its next code: Primary, Alt 1, Alt 2…, Carton, then back to Primary. |

> **Note:** The chosen code is used for both the barcode and the printed UPC number on the label. Choices reset to the primary UPC when you press Reload viewtouch.db or restart the program.

# 6. Screen 4 — Preview & Print

The large area shows the label exactly as it will print, for one item at a time.




Screen 4 — toolbar (wraps onto a second row on the Pi's screen) and the label preview

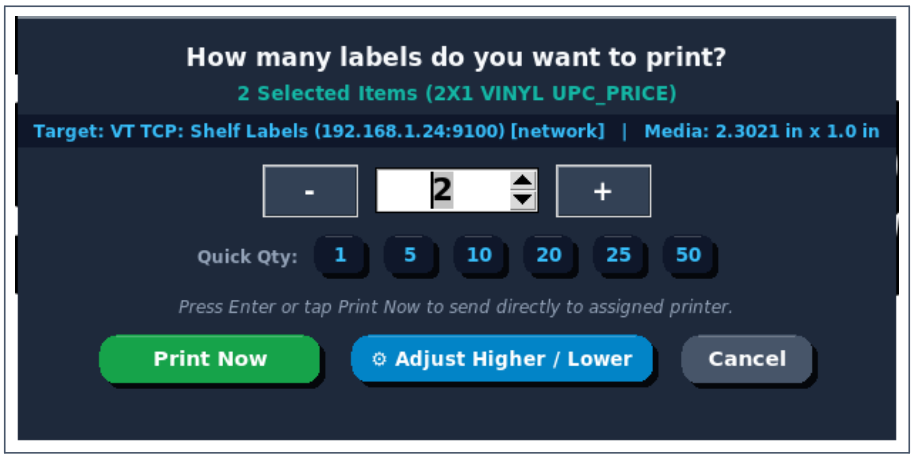
| Control | What it does |
|---|---|
| **Template** | Chooses which label design to use. The preview updates immediately. |
| **Print Labels** | Opens the print window (section 7). |
| **Arrow buttons and 1/5** | Steps through the ticked items. The number shows which item is previewed. |
| **UPC** | Lists every code for the previewed item (e.g. *Alt UPC 1: 04252614*). Pick one to print that code on this item's label. |
| **Edit Report** | Opens the label designer for the current template (section 8). |
| **Create Report** | Makes a new template as a copy of the current one (section 8.6). |
| **Printer Offset** | Printer position, darkness, resolution and calibration for this template (section 7.3). |
| **Printer badge** | Shows which printer and label size this template uses. |
| **Double-click the preview** | Switches between the sharp preview and the exact printer-dot view. The caption under the label says which one you are seeing. |

# 7. Printing labels

## 7.1 Step by step

1. On screen 1 choose how to find your items (e.g. **Retail Items**), or use **Tree View** and pick a brand.
2. On screen 3 tick the items. Use **Print UPC** or **Next UPC** if a label needs an alternate or carton UPC.
3. Press **Next >>**. On screen 4 choose the **Template** (e.g. *2X1 VINYL UPC_PRICE*).
4. Check the preview. Use the arrow buttons either side of the item counter to look at each item.
5. Press **Print Labels**. If the template has a Variable field, you are first asked for the text (press OK, or Skip).
6. Enter **how many labels** per item and press **Print Now** (or Enter). A green message confirms the labels were sent.

## 7.2 The print window




The print window

| Control | What it does |
|---|---|
| **How many labels?** | Number of labels printed **for each selected item**. Use - and +, type a number, or tap a Quick Qty button (1, 5, 10, 20, 25, 50). It starts at the first item's Print Qty. |
| **Target / Media line** | The printer and label size the job will go to. Check this before printing. |
| **Print Now** | Sends the labels. Pressing Enter does the same. |
| **Adjust Higher / Lower** | Closes this window and opens Printer Offset (section 7.3). |
| **Cancel** | Closes without printing. |

> **Example:** 3 items ticked and 2 entered here prints 6 labels — 2 of each item.

### Where labels go

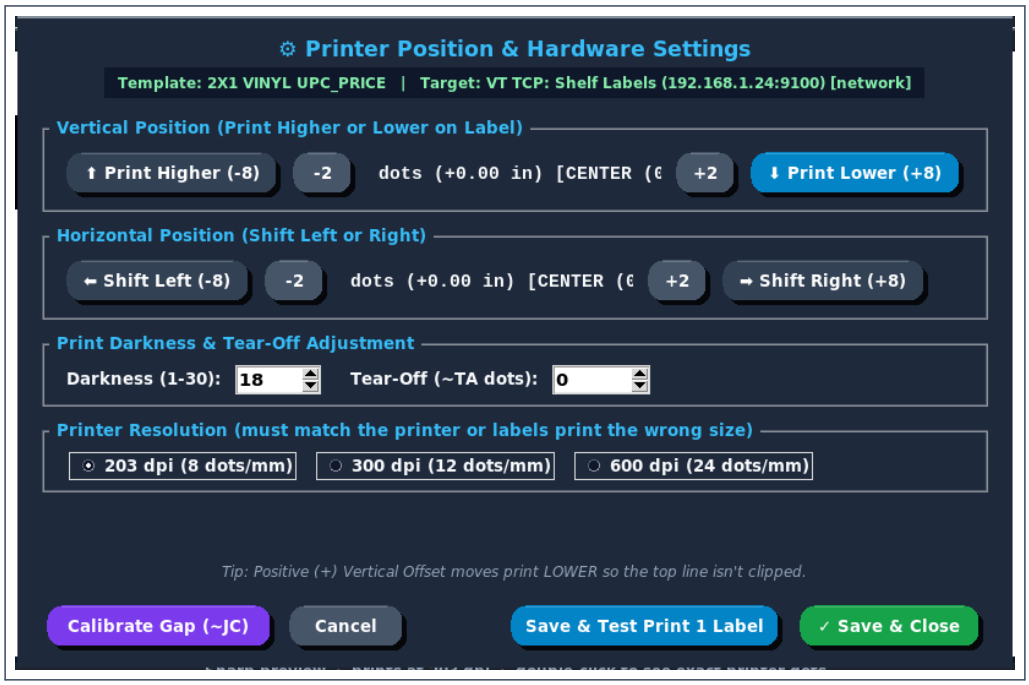
Each template remembers its printer. The printers come from ViewTouch's printer settings.

| Printer type | How the label is sent |
|---|---|
| **Network label printer (e.g. Zebra Large Labels, Shelf Labels, Stick Labels)** | Sent directly to the printer's IP address as a ZPL picture. |
| **CUPS printer queue** | Label printers get the same ZPL picture; ordinary printers get a PDF page the size of the label. |
| **PDF Preview (Desktop)** | Opens a PDF of the labels on screen instead of printing. |
| **Receipt printers (Epson, receipt, kitchen…)** | Get a plain text version. Don't use these for labels. |

**Changing a printer's IP:** Change it in ViewTouch's printer settings. Templates follow the printer by name, so a template set to **Stick Labels** automatically uses Stick Labels' new address.

## 7.3 Printer Offset (printer settings)

Open with **Printer Offset** on screen 4 or **Adjust Higher / Lower** in the print window. Settings are saved **per template**.




Printer Offset — printer position, darkness, resolution and calibration

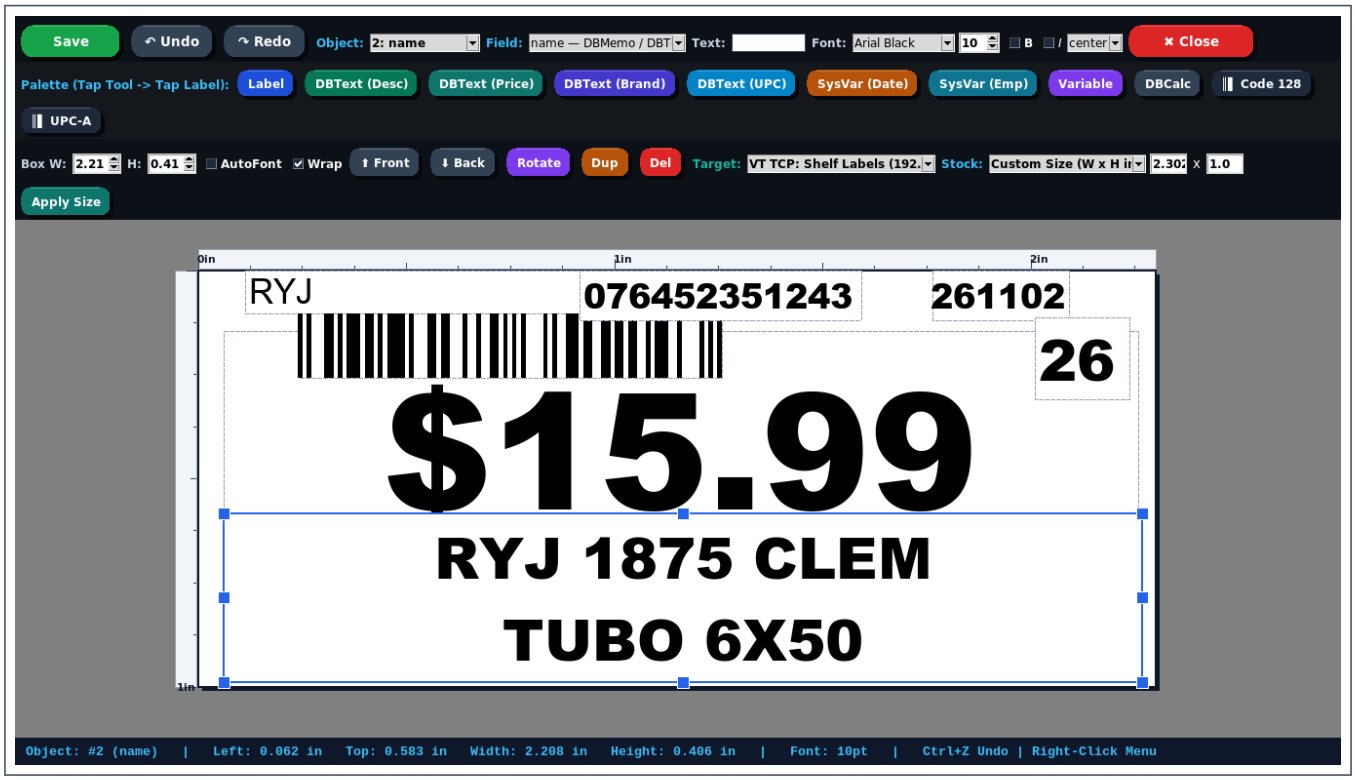
| Setting | What it does |
|---|---|
| **Vertical Position** | **Print Higher (-8)** / **Print Lower (+8)** move everything up or down by 8 dots (about 1 mm). -2 / +2 make finer moves. Use this if the top or bottom line is cut off. |
| **Horizontal Position** | **Shift Left** / **Shift Right** move everything sideways, the same way. |
| **Darkness (1–30)** | Higher is darker. Raise it if bars or text look faint; lower it if bars bleed together. |
| **Tear-Off** | Adjusts where the label stops for tearing off. Leave at 0 unless labels tear in the wrong place. |

| Setting | What it does |
|---|---|
| **Printer Resolution** | 203, 300 or 600 dpi. **Must match the printer** (shown on the printer's label) or labels print the wrong size. |
| **Save & Close** | Saves and closes. |
| **Save & Test Print 1 Label** | Saves and prints one label of the previewed item, so you can check the position. |
| **Calibrate Gap** | Makes the printer feed a few labels to measure the label length. Do this after loading a new roll or a different label size. |
| **Cancel** | Closes without saving. |

**Note:** Test Print and Calibrate Gap only work with network printers (ones with an IP address).

# 8. Editing labels (the label designer)

Press **Edit Report** on screen 4. The designer opens the current template, filled in with the item being previewed, so you can see real data while you work.

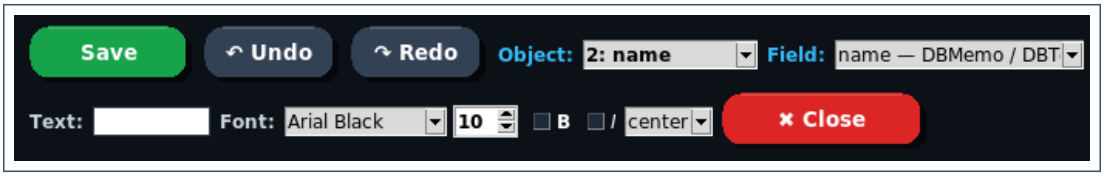



The label designer — three toolbars above the label; the description is selected (blue box and handles)

## 8.1 Moving and sizing pieces with the mouse or touch

- **Click** a piece (description, price, barcode…) to select it. A blue dashed box with 8 handles appears.
- **Drag** the piece to move it. **Drag a handle** to change its box size.
- **Ctrl-click** or **Shift-click** to select several pieces; **drag on an empty area** to draw a selection box around them.
- **Double-click** a piece for its Properties window (field, text, exact position, rotation).
- **Right-click** for a menu: Add Field, Bring to Front, Send to Back, Position / Properties, Duplicate, Rotate 90°, Delete Component.

## 8.2 Top toolbar




Top toolbar (shown in two parts)

| Control | What it does |
|---|---|
| **Save** | Saves the template and closes the designer. |
| **Undo / Redo** | Steps back or forward through your changes. |
| **Object** | Picks a piece by name (useful when pieces overlap). |
| **Field** | Changes what the selected piece shows (see the field list in 8.5). |
| **Text** | The words for a static Label or Variable piece. Press Enter to apply. |
| **Font / size / B / I** | Font (Sans, Serif, Mono, Arial, Arial Black, Courier New), point size, bold, italic. |
| **Alignment** | left, center or right inside the piece's box. |
| **Close** | Closes the designer. If there are unsaved changes it asks: **Yes, Save**, **No, Discard** or **Cancel**. |

## 8.3 Palette — adding new pieces

Tap a palette button, then tap on the label where the new piece should go. Press **Esc** to cancel.




Palette (shown row by row)

| Palette button | Adds |
|---|---|
| **Label** | Fixed text you type (e.g. SALE). |
| **DBText (Desc) / (Price) / (Brand) / (UPC)** | The item's description, price, brand or UPC number. |
| **SysVar (Date) / SysVar (Emp)** | Today's date (YYMMDD) or the employee's name as F LAST. |
| **Variable** | Text you are asked for each time you print (e.g. RECEIVED). |
| **DBCalc** | Pack calculation, e.g. PK:10 ($1.20/EA). |
| **Code 128 / UPC-A** | A barcode of the item's UPC (the one chosen on screen 3 or 4). |

## 8.4 Second toolbar




Second toolbar (shown row by row)

| Control | What it does |
|---|---|
| **Box W / H** | Exact width and height of the selected piece, in inches. |
| **AutoFont** | On: text shrinks or grows to fill its box. Off: uses the exact font size you set (the ILSA labels use exact sizes). |
| **Wrap** | Lets long text use two lines. |
| **Front / Back** | Brings the piece in front of, or sends it behind, overlapping pieces. |
| **Rotate** | Turns the piece 90°. |
| **Dup / Del** | Duplicates or deletes the selected piece. |
| **Target** | The printer this template prints to. |
| **Stock** | Picks a standard label size (2.0×1.0, 2.5×0.75, 3.0×2.0, 1.5×1.0, 2.25×1.25) or Custom. |
| **W × H + Apply Size** | Type an exact label size in inches and press Apply Size. |

### Keyboard shortcuts

| Keys | Action | Keys | Action |
|---|---|---|---|
| **Ctrl+Z / Ctrl+Y** | Undo / Redo | Arrow keys | Nudge 0.01 in |
| **Ctrl+C / X / V** | Copy / Cut / Paste | Shift+Arrows | Nudge 0.05 in |
| **Ctrl+D** | Duplicate | R | Rotate 90° |
| **Ctrl+A** | Select all pieces | Delete | Delete selected |
| **Esc** | Cancel palette tool | | |

### 8.5 Field list

| Field | Prints |
|---|---|
| **Label (Static Text)** | The text you type. |
| **Variable** | Text asked for when printing (default RECEIVED). |
| **Date (YYMMDD) / Date (MM/DD/YYYY) / Year (YY)** | Today's date in that format. |
| **Employee (F LAST)** | The active ViewTouch employee's name, e.g. P SMITH. |
| **Description, Retail Price, Manufacturer / Brand, Category, UPC / PLU** | The item's details from ViewTouch. |
| **DBCalc** | Pack quantity and per-unit price. |
| **Code 128 / UPC-A / UPC-E / Code 39** | A barcode. UPC fields automatically use the right type for the code chosen (UPC-A, UPC-E, EAN-13), and the check digit is always calculated correctly. |

**ILSA labels:** Labels imported from ILSA keep ILSA's bar width, the digits printed under the bars, and their exact text placement. These settings are carried over automatically and are not shown in the designer.

### 8.6 Creating a new label (Create Report)

1. On screen 4 choose the template closest to what you want, then press **Create Report**.
2. Enter a **Label / Store Name** (this becomes the template name).
3. Choose the **Target Printer** and the **Media Stock / Size**.
4. Press **Create & Open Designer**. The new template is a copy of the one you started from, ready to edit.

## 9. Importing labels

Use **Lookup Template** on screen 1. A file window opens; choose the file and press Open.

### ILSA labels (.rtm)

Label designs saved from the old ILSA system (ReportBuilder **.rtm** files) import directly. The program reads the label size, the position and size of every piece, fonts, alignment, barcode type, bar width and the digits under the barcode, and creates a template with the same name as the file.

1. Copy the .rtm file to the Pi (a USB stick, or put it in the folder that **Open Template Folder** opens).
2. Press **Lookup Template** and choose the file.
3. If a template with that name already exists you are asked whether to replace it.
4. A message confirms the import. The new label opens in the preview.
5. Open **Printer Offset**: set the right printer, check the resolution, and print a test label.

Fields understood in ILSA files: description (PLU_Desc), UPC (PLU_Num), price, manufacturer, category, dates (yymmdd, yy, mm/dd/yyyy), fixed text labels, and UPC-A, UPC-E, EAN-13, Code 39 and Code 128 barcodes.

**After importing:** ILSA saved some labels a little wider than the actual stock (for example 2.3 in for a 2 in label). This is kept as-is so the layout matches ILSA. If the right edge is cut off, adjust the size with W × H + Apply Size in the designer.

### Other files

| File | Result |
|---|---|
| **.json** | Template file(s) from this program or the other ViewTouch label programs. Every template in the file is added. |
| **.prn, .nlbl, .slnx** | ZebraDesigner / NiceLabel / BarTender files can't be read. A new template with the file's name is created as a copy of RECEIVING LABEL for you to design in the designer. |

# 10. Your label templates

These four templates are installed. They match the labels from the ILSA system.

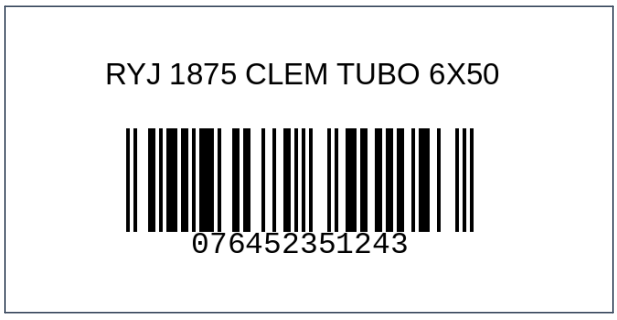
**2X1 VINYL UPC_PRICE — shelf price label**

Brand, UPC, date (YYMMDD), barcode, year code, large price in Arial Black, and description. Printer: Shelf Labels.



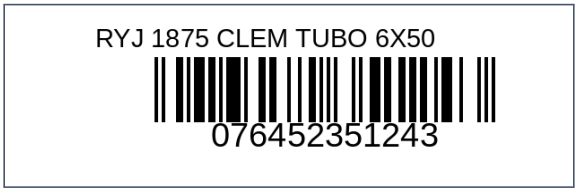

**2X1 VINYL UPC_DESC — description and barcode**

Description with a full-size UPC-A barcode and the digits underneath. Printer: Zebra Large Labels.



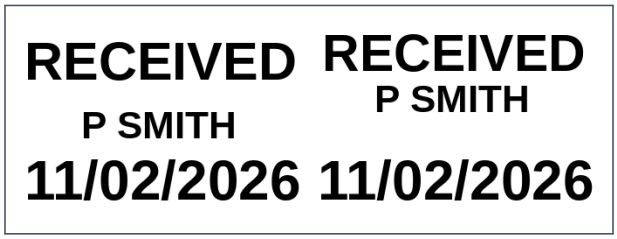
### CIGAR STICK LABEL UPC_DESC — 2.0 × 0.56 in

Description and UPC-A barcode for cigar sticks. Printer: Stick Labels (the cigar printer at 192.168.1.25).




### RECEIVING LABEL — 2.2 × 0.75 in, two across

RECEIVED, the employee's name and today's date, twice side by side. Needs no item.




**Back up your templates:** Your adjusted templates live in `~/.local/share/ViewTouch/claude_label_templates.json`. Copy that file somewhere safe after making changes you want to keep, or run the upload command in section 12.

# 11. Updating the program

New versions are published to your GitHub repository. To update, open a Terminal on the Pi and run:

```
cd ~/viewtouch-main
git pull myfork claude-label-utility
```

Then close and reopen the label utility. Your templates and printer settings are kept.

If a new version needs new fonts or packages, also run the installer (it can be run any time):

```
bash ~/viewtouch-main/modern/tools/install_label_utility.sh
```

The installer sets up the launcher ViewTouch's Label Utility button uses, installs any missing Python parts, and installs Microsoft's Arial, Arial Black and Courier New fonts used by the ILSA labels. It may ask for the Pi's password.

# 12. Troubleshooting

| Problem | What to do |
|---|---|
| **Print says it was sent, but nothing prints** | Check the Target line names the right printer. Then test the printer itself from a Terminal (replace the IP):<br>`printf '^XA^FO40,40^A0N,50,50^FDTEST^FS^XZ' > /dev/tcp/192.168.1.24/9100`<br>If TEST doesn't print, the printer is paused, out of labels/ribbon, open, or off. |
| **Labels go to the wrong printer** | Open Printer Offset or the designer and check Target. Check the printer's IP in ViewTouch's printer settings. |
| **Top or bottom line cut off** | Printer Offset: Print Higher or Print Lower, then Save & Test Print. |
| **Right edge cut off** | The template is wider than the labels. Reduce the label width in the designer (W × H, Apply Size). |
| **Labels print too small or too large** | Printer Resolution in Printer Offset doesn't match the printer (203 vs 300 dpi). |
| **Labels skip, or print across two labels** | New roll or different size: Printer Offset, Calibrate Gap. |
| **Faint print, or bars run together** | Raise or lower Darkness in Printer Offset. |
| **Barcode won't scan** | Check the right UPC was chosen (screen 3, UPC on Label). Make sure the barcode isn't cut off and Darkness is right. |
| **Big price touches the barcode** | The Arial Black font isn't installed. Run the installer (section 11). |
| **An item is missing or has an old price** | Press Reload viewtouch.db. Items marked inactive in ViewTouch, and restaurant menu items, are not listed. |
| **Program won't open or closes by itself** | Send the log (below) for diagnosis. |

### Sending the log for help

Right after a problem (before restarting the Pi), run:

```
bash ~/viewtouch-main/modern/tools/push_label_log.sh
```

This uploads the error log, the last label sent to a printer, and your current templates to the **pi-logs** branch of your GitHub repository, where they can be reviewed. It doesn't change any of your files.

# 13. Quick reference

| To… | Do this |
|---|---|
| **Open the label utility** | ViewTouch Manager screen > Label Utility |
| **Print shelf labels for some items** | Retail Items > tick items > Next >> > Template > Print Labels > quantity > Print Now |
| **Print labels for one brand** | Tree View > double-click the brand > tick items > Next >> > Print Labels |
| **Print receiving labels** | Screen 1: double-click RECEIVING LABEL > Print Labels > quantity > Print Now |
| **Print a carton or alternate UPC** | Screen 3: Print UPC or Next UPC — or screen 4: UPC drop-down |
| **Find an item by any of its UPCs** | Screen 3: type or scan the code into Quick Search |
| **Move the print up or down** | Screen 4: Printer Offset > Print Higher / Lower > Save & Test Print |
| **Set up a new label roll** | Printer Offset > Calibrate Gap |
| **Change a label's layout** | Screen 4: Edit Report > drag pieces > Save |
| **Make a new label from an existing one** | Screen 4: Create Report > name, printer, size > Create & Open Designer |
| **Bring in an ILSA label** | Screen 1: Lookup Template > choose the .rtm file |
| **See exact printer dots** | Screen 4: double-click the preview |
| **Update the program** | `cd ~/viewtouch-main && git pull myfork claude-label-utility` |
| **Send logs for help** | `bash ~/viewtouch-main/modern/tools/push_label_log.sh` |